EncoderMoE: Classifying User-Defined Gestures Using MoE Encoders and Meta-Learning

(/pdf?id=B6K7b94VvS)

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04 May 2026 (modified: 27 May 2026) NeurIPS 2026 Conference Submission
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
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TL;DR: We combine meta-learning with a Mixture-of-Experts encoder to enable few-shot cross-subject classification of user-defined gestures from EMG-IMU signals in a cohort of users with upper-body motor impairments.

Abstract:
Wearable gesture interfaces using surface electromyography (sEMG) and inertial measurement units (IMUs) offer compelling pathways for accessible human-computer interaction, yet generalizing to novel users remains a fundamental challenge. This problem is especially pronounced for user-defined gesture systems, where physiological heterogeneity across individuals is compounded by semantic heterogeneity: different users may perform entirely different motions for the same gesture label. This challenge is further amplified for users with motor impairments, whose signal characteristics diverge substantially from those of unimpaired users on which most prior work is trained. We formulate user-defined multimodal gesture recognition as a few-shot cross-subject classification problem and propose \textbf{EncoderMoE}, combining Model-Agnostic Meta-Learning (MAML++) with an encoder-level Mixture-of-Experts (MoE) module. MAML++ learns parameter initializations that adapt to a new user from as few as one labeled example per gesture; encoder-level MoE provides representational diversity across specialized sub-encoders to capture heterogeneous physiological and semantic structure across diverse disability profiles. We evaluated 32 participants, 26 with upper-body motor impairments, performing user-defined executions of 10 gesture prompts via a 28-sensor EMG-IMU setup. EncoderMoE achieves 86.7% accuracy on 1-shot 3-way classification (chance: 33.3%), outperforming subject-specific transfer learning by 14.5 points. Ablation studies confirm that meta-learning and representational specialization are jointly necessary, with parameter-matched controls ruling out model capacity as a confound. These results demonstrate that pairing fast adaptation with heterogeneity-aware representations is an effective strategy for personalized biosignal interfaces for users with motor impairments.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Financial Support: Kai Malcolm (/profile?id=~Kai_Malcolm1)
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: Generalization and multi-task learning (e.g., transfer learning, meta-learning, OOD generalization)
Secondary Area: Neuroscience and cognitive science (e.g., neural coding, brain-computer interfaces)
Contribution Type: General: Most submissions will fall into this type.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Editing (e.g., grammar, spelling, word choice), Drafting sections of the paper, Understanding technical concepts, Implementing standard methods
LLM Experiment: Opt in to include this paper in the LLM-assisted peer review experiment.
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: Momona Yamagami (/profile?id=~Momona_Yamagami1)
Submission Number: 31103

Discussion (?id=B6K7b94VvS#discussion)

Filter by reply type... Filter by author... Search keywords... Sort: Newest First

Everyone Program Chairs Submission31103... Submission31103... Submission31103... *24 / 24 replies shown*
Submission31103... Submission31103... Submission31103... Submission31103...
Submission31103... Submission31103...

Add: **Withdrawal**

Citation Checker by Program Chairs

Citation Checker by Program Chairs 04 Aug 2026, 13:19 (modified: 05 Sept 2026, 00:20)
Program Chairs, Senior Area Chairs, Area Chairs, Authors Revisions (/revisions?id=SVt9eejOeQ)

Comment:

Automatically flagged references

Here we list the references that were automatically flagged by our checker. Note that our checker may miss many hallucinated references, and some of the listed references may have been flagged erroneously. **It is therefore crucial that you check them manually, following the guidance above. We rely on the Area Chairs to decide whether these constitute violations of responsible authorship.**

For papers falling into either of the following categories, we expect your default recommendation to be a rejection on the grounds of irresponsible authorship. This is strong presumption rather than a strict rule; as the Area Chair, you retain discretion, and may recommend otherwise where the specific circumstances of a paper warrant it:

1. The paper contains at least two hallucinated references according to the criteria described below the report.

2. The paper contains one hallucinated reference and there is some additional evidence of AI slop.

If a paper has only one problematic reference, but no additional concerns about the quality of the paper or its presentation, please simply alert the authors in your meta-review.

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Citation Checker Report

Reference Check Summary

0/29 problematic references.

Hallucinated/problematic reference criteria

These are adopted from ICLR and ICML 2026 hallucinated reference guidance:

- If you are unable to find any document or publication with a title anywhere near the one provided in the reference, it should be considered hallucinated. Minor differences in title should not be considered evidence of a hallucinated reference.
- If the reference's authors are only slightly incorrect, e.g., a few subtle misspellings, missing a few authors, or wrong authorship order, we do not consider this to imply a hallucinated reference, but please let the submission's authors know in your meta-review that they should fix this issue.
- If the author list has additional issues, e.g., added authors that are not actually authors of that publication or missing a large fraction of authors, then the reference should be considered hallucinated.
- If the reference's venue is a real venue but is incorrect for that reference, we do not consider this to imply a hallucinated reference, but please tell the submission's authors in your meta-review, these include wrong journal name, arXiv ID, etc.
- If there is no evidence that the reference's venue exists, the reference should be considered hallucinated.
- Finally, we note that there are many generic references that refer to existing artifacts (e.g. "GPT-4o, OpenAI et al. 2024") but do not refer to any real document. These are fine.
- Even with the above guidelines in mind, there will likely be some borderline cases. Please bring those up with your SAC if you need additional guidance.

General Response

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

30 Jul 2026, 11:57 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Reviewers Submitted, Authors

Comment:

We thank the reviewers for their careful assessment of our work, and appreciate that the reviewers found the paper to demonstrate “strong results in adapting to esoteric user behaviors” (Reviewer ARaC) and that the “combination of MAML++ and MoE at the encoder level is well-motivated and empirically demonstrated to be necessary for high performance in the 1-shot regime” (Reviewer Lpuc). Below, we provide our responses to each individual reviewer in the appropriate thread. We note that some of the requested experiments are still running: items that depend on missing numbers are marked with [TODO], and we will follow up in the replies when those runs finish. For a detailed response on comparing our work to MoEMeta, see our answer to reviewer ARaC.

Summary: Aggregating revisions across all threads

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:31

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Reviewers Submitted, Authors

Comment:

Summary of revisions and new results

We thank the reviewers for a discussion that has strengthened our paper. Several requests were raised in multiple threads, so we summarize the revisions and completed experiments here; per-reviewer details follow in the individual threads.

1. Revised contributions. No component of EncoderMoE is novel in isolation. We have added and discussed MoEMeta [Wu et al., 2025], and relabeled Contribution 1 from a method to an application contribution:

> *Contribution 1. Application.* EncoderMoE: an encoder-level mixture of experts over multimodal EMG–IMU signal whose experts and routing gate are both meta-learned and adapted from a single example per gesture. No component is novel in isolation; the contribution is the instantiation, motivated by an inductive-bias argument explaining why routed encoder capacity suits the user-defined gesture regime, in which the label set is shared across users but no global gesture-to-label mapping exists.

> *Contribution 2. Empirical analysis.* Parameter-matched ablations against subject-specific and cross-subject baselines, and against reduced variants of EncoderMoE itself, showing that both components (an encoder-level mixture of experts, and inner-loop adaptation over that mixture) are required in this regime.

2. MoEMeta comparison. MoEMeta cannot be run directly on sensor data, so we ported its two substantive design decisions onto our data with everything else fixed. *Port A* (support-derived vs. query-conditioned routing): 91.24% vs. 91.25%, a null, on the basis of which we **withdrew** the accuracy claim for query-conditioned routing. *Port B* (frozen vs. adapted mixture): restricting inner-loop adaptation to the classification head, 0.15% of parameters, gives 59.76% ± 2.98% against 88.4%, a 28.6-point cost across all four held-out subjects. Both interpretations were stated before the runs.

3. Kaifosh baseline, corrected. Our published evaluation of [11] supplied signal-normalized input to a decoder expecting noise-floor-normalized input, placing its compression nonlinearity in the near-linear regime. Re-run with their preprocessing (40 Hz high-pass, noise-floor normalization): linear probing **66.5%** (published 62.1%), full fine-tuning **49.2%** (published 56.0%). We report both directions and are softening L289–290, which leans on the size of the gap.

4. Modality ablation, and a claim we are withdrawing. Zero-masking each modality, fixed 24/4/4 split, 1-shot, four held-out subjects, 500 episodes:

	3-way	5-way	10-way
Fused	87.6	80.3	63.3
EMG-only	88.7	71.8	45.2
IMU-only	90.4	81.5	65.4

IMU is the dominant modality here, and EMG-only degrades sharply as the label set grows (−18.1 pp vs. fused at 10-way, Holm-adjusted p = .023). IMU-only is statistically indistinguishable from fused at every vocabulary size. **We are removing the claim at L34–38 that hybrid EMG-IMU interfaces overcome complementary failure modes**, and we do not claim fusion improves over IMU alone on this split.

5. Meta-training user sweep. N ∈ {4, 8, 16} meta-training users, fixed val/test split, 3 subsample seeds per N: 82.4% ± 0.5%, 85.3% ± 1.7%, 87.9% ± 0.5%. The gain continues from N=8 to N=16, past the point where within-subject sample scarcity alone would explain the cross-subject advantage.

6. Metric-learning baseline. Prototypical Networks on our meta-trained backbone (identical encoder, nearest-prototype readout replacing meta-learned adaptation), 1-shot: 78.7 / 71.7 / 52.2 at 3-, 5- and 10-way, against 88.4 / 78.2 / 67.7 for EncoderMoE.

7. One correction to the record. The meta-review states that "performance on 3-way classification (67.7%) is notably lower than 10-way classification (86.7%)." These are transposed: EncoderMoE achieves **86.7% at 1-shot 3-way** and **67.7% at 1-shot 10-way**. The meta-review's underlying point, that large vocabularies remain the harder regime, is correct and we state it as a limitation; we flag the numbers only so the record is accurate.

8. Baselines. Table 1 reports two comparisons against published methods, which we are relabeling to make explicit: Kaifosh and Reardon [11] (Nature 2025, pretrained on 4,800+ participants) and Côté-Allard-style transfer learning [7], the row currently labeled "No MAML, No MoE". For [7], we pretrain on our own users rather than their forearm-band users, whose sensor configuration is not incompatible with ours (num channels, no IMU, etc).

9. Not completed. The ARaC upper bound, the transformer comparison, and EncoderMoE trained on the 2 kHz EMG-only inputs did not finish within the discussion period; all three are committed for camera-ready.

Official Review of Submission31103 by Reviewer Lpuc

Official Review by Reviewer Lpuc 27 Jun 2026, 12:31 (modified: 14 Aug 2026, 03:40)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Lpuc

Revisions (/revisions?id=upI2ISSOwU)

Summary:
The paper presents EncoderMOE, a method that combines mixture of experts (MoE) and model-agnostic meta learning (MAML), for few-shot classification of user-defined gestures from EMG and IMU signals.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths:

- This work addresses the central challenge of recognizing user-defined gestures in highly heterogeneous populations, particularly users with motor impairments.
- The combination of Model-Agnostic Meta-Learning (MAML++) and Mixture-of-Experts (MoE) at the encoder level is well-motivated and empirically demonstrated to be necessary for high performance in the 1-shot regime.
- The model's routing mechanism is shown to discover latent ability-level structure in the data, even without explicit disability labels, indicating that the approach captures meaningful user variation.
- The methodological details are thoroughly stated supporting reproducibility

Weaknesses:

- Limited baseline evaluations especially architecture baselines: All the reported baselines are limited to the architecture proposed in the current paper. How does the method compare to state-of-the-art in EMG-IMU gesture recognition is not shown.
- The architectural choice is not empirically validated. The authors argue that they prefer LSTMs over attention-based temporal models because the recurrent inductive bias reduces the degrees of freedom MAML's inner loop of 1-shot adaptation. But this is not theoretically or empirically validated. Also the choice of mixture of CNN encoders is not compared against, for example, a single or mixture of transformer encoders.
- Linear probing and finetuning results are reported on a completely different dataset with EMG-only inputs and information on number of classes is missing. It is not possible to compare these results with the proposed method.
- Single dataset evaluation: although this is explicitly stated as a limitation, evaluation on a single dataset with 32 subjects limits the extent to which the claims can be generalized.

Quality: 3: good
Clarity: 3: good
Significance: 2: not good
Originality: 3: good
Questions:

- How is the choice of 22 experts made? The hyperparameter optimization shows that peak accuracy is also reached at N=8.
- Did you do an ablation study on the inputs? What is the contribution of EMG vs IMU to gesture classification?
- Have you done a subject-level (not ability level) routing analysis? Since number of experts is not very small compared to number of subjects, how do you make sure that there is no single expert for atleast some of the subjects?

Limitations:
yes

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
no

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Response to Lpuc

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

30 Jul 2026, 12:06

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Lpuc

Comment:

0. State-of-the-art comparison, number of classes.

We compare against Kaifosh [11] and Cote-Allard style transfer learning [7]. We will revise the paper to make these more explicit state-of-the-art comparisons. As per our response below to the EMG vs IMU contribution, we will run an EMG-only condition for EncoderMoE, and make the K-shot N-way comparisons to [11] more explicit.

1. Why 22 experts when the sweep also peaks near N=8?

These are two different procedures and we should have been more clear. E=22 came out of the hyperparameter search (App. A.2), which varied **all hyperparameters jointly**. The expert-count sweep in Figure 3 came later and varies only E with **everything else fixed at the chosen configuration** (i.e., the hyperparameters fit for E=22). App. A.4 already notes that within-trial inter-subject variability exceeds the differences between neighboring expert counts, so E=8 and E=22 are not distinguishable on this evidence.

The practical implication favors Reviewer Lpuc's reading: E=8 gives comparable accuracy at roughly a third of the expert-bank parameters, and we will state this in A.4 rather than leaving E=22 the only optimum. We kept E=22 for the reported results because it is the configuration found by hyperparameter tuning, against which all ablations were matched.

2. Subject-level routing; is any subject served by a single expert?

No subject has a dedicated expert (prevented by the architecture). We will add the following to the appendix.

"Method. For all 32 subjects we collected per-sample gate vectors over 50 episodes each (1,350 query samples per subject, equal counts by construction) in the post-adaptation configuration, and reduced each subject to a mean gate vector over the 22 experts. The direct test for a private expert is a margin: for each subject, the largest amount by which its mean weight on an expert exceeds the highest weight any other subject places on that same expert. A subject with a dedicated expert produces a margin approaching 1.0.

Results. The largest margin across all 32 subjects is 0.050 and the median is -0.006, so for more than half the subjects there is no expert on which they hold the highest mean weight. The largest mean weight any subject places on a single expert is 0.233, against 0.111 for uniform allocation over the k=9 active experts. No expert draws more than 0.156 of its total gate mass from any one subject, and every expert is used by at least 30 of the 32 subjects. Pairwise cosine similarity between subject mean gate vectors averages 0.86, where a per-subject expert assignment would drive it toward zero. The eight held-out subjects give the same picture (largest margin 0.040, mean cosine 0.83)."

Routing is also constrained architecturally: with top-k = 9 of 22, exactly nine experts are active on every sample, so a one-expert-per-subject assignment is not representable. Per-sample gate entropy is 2.158 nats against a ceiling of ln(9) = 2.197 over the active set, so weight is close to uniform across the experts each sample activates.

3. EMG vs. IMU contribution (also raised by Reviewers xh89 and fkxS)

We will add the additional evaluation to Experiments; evaluation results will be updated when results come in [TODO].

Here, we zero-mask the channels of each modality separately, and run all three cases (EMG-only, IMU-zero-masked; IMU-only, EMG-zero-masked; both modalities, no zero masking). We are adding a modality ablation on the fixed 24/4/4 split at 1-shot 3-way: fused 88.4%, EMG-only [TODO], IMU-only [TODO].

4. Architecture choices are not validated against Transformer alternatives; novelty concerns

On validation: the original choice came from pilot testing and the reasoning at lines 171-173, which is a hypothesis rather than evidence. We are replacing the LSTM and CNN components with Transformer-based equivalents and will report the comparison. [TODO].

We accept that no individual component is new; we will reframe our contribution because of MoEMeta as seen above. We claim the instantiation and the empirical result: in this specific regime (one labeled example per class, user-defined label semantics, high physiological heterogeneity from users with motor impairments), routed encoder capacity and meta-learned adaptation are jointly necessary, with parameter-matched controls ruling out capacity as the explanation, and routing's benefit concentrates at K=1 and largely disappears by K=5. Contribution 1 has been rewritten to claim that rather than architectural novelty.

"Method. EncoderMoE: an encoder-level mixture of experts over multimodal EMG–IMU signal whose experts and routing gate are both meta-learned and adapted from a single example per gesture. No individual component is novel; the contribution is the instantiation, motivated by an inductive-bias argument for why routed encoder capacity suits the user-defined gesture regime where gesture labels are shared across users but their execution is not."

Response to Rebuttal

Official Comment by Reviewer Lpuc 02 Aug 2026, 11:55

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Lpuc

Comment:

I thank the authors for the brief rebuttal. The assessment of the paper has not changed as all the requested changes are only answered as "will be done", but I do not see any results of any additional experimental results presented during rebuttal period. All my concerns listed in the weaknesses section remain as is.

Replying to Response to Rebuttal

Response to Lpuc (1/2)

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:41

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Lpuc

Comment:

Response to the 02 Aug comment: completed results (1/2)

The reviewer is right that our earlier reply was largely commitments. Below are completed results, and a correction to a factual point in W3 that our own wording caused.

W3: the Kaifosh rows are not a different dataset, and they are 3-way. This is our fault for writing "dataset variant" at L230. Those rows are evaluated on **the same 32 participants as every other row in Table 1**, under the **same episodic protocol: 1-shot, 3-way, 9 query samples per class, 500 episodes per test participant, Leave-Two-Subjects-Out**. The only difference is input format: their model requires 2,000 Hz raw EMG and does not accept IMU, so it is evaluated on a non-downsampled EMG-only version of our own data rather than the 20 Hz multimodal windows. Their pretrained 9-class head is replaced with a fresh 3-way head, identically to every other non-meta-learned condition. We will state N-way, K-shot, episode count and input format explicitly in the caption and in §4.2.

W3, continued: corrected Kaifosh evaluation. We also found a defect in our own evaluation of their model. Their decoder expects input normalized so the *noise* floor has unit standard deviation; we supplied *signal*-normalized input (to match EncoderMoE's preprocessing). Re-run with their preprocessing applied (40 Hz high-pass, noise-floor normalization): linear probing **66.5%** (published 62.1%), full fine-tuning **49.2%** (published 56.0%). We report both directions and are softening the sentence at L289–290 that leans on the size of the gap.

Q2: modality ablation. Channels are zero-masked rather than deleted, so input width and parameter count are unchanged. Fixed 24/4/4 split, 1-shot, test accuracy over four held-out subjects, 500 episodes:

	3-way	5-way	10-way
Fused	87.6	80.3	63.3
EMG-only	88.7	71.8	45.2
IMU-only	90.4	81.5	65.4

The fused control reproduces the fixed-split number in Table 4 (88.4 ± 2.9), confirming the masking harness is inert. A two-way repeated-measures ANOVA gives a significant condition × n-way interaction ($F(4,12) = 19.2$, $p < .001$). EMG-only falls significantly below fused at 10-way ($\Delta = -18.1$ pp, Holm-adjusted $p = .023$, all four subjects agreeing) and marginally at 5-way ($\Delta = -8.5$ pp, adjusted $p = .076$); IMU-only is indistinguishable from fused at every vocabulary size (adjusted $p = .385$ / .631 / .646).

We report this straight: IMU is the dominant modality here and EMG-only degrades sharply as the label set grows, but **we do not claim fusion improves over IMU alone on this split**, and we are removing the claim at L34–38 accordingly. With four test subjects the fusion contrast is underpowered (the smallest detectable effect at 80% power is roughly 6–8 pp), and we will re-run this under L2SO for camera-ready so it enters the same paired design as our main results.

Continued in the following comment: metric-learning baseline, architecture validation, and results posted in other threads.

Replying to Response to Lpuc (1/2)

Response to reviewer Lpuc (2/2)

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:46

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Lpuc

Comment:

Response to the 02 Aug comment: completed results (2/2)

W1: metric-learning baseline. We have evaluated Prototypical Networks on our own meta-trained backbone: identical encoder and identical support budget, with the meta-learned inner-loop adaptation replaced by a nearest-prototype readout. This is the parameter- and backbone-matched comparison, and it tests rather than assumes the claim at L110–112. 1-shot:

n-way	ProtoNet (encoded)	EncoderMoE
3-way	78.7 ± 14.0	88.4
5-way	71.7 ± 12.9	78.2
10-way	52.2 ± 10.8	67.7

The prototype readout is below EncoderMoE at every vocabulary size and the gap widens with N (9.7, 6.5, 15.5 points), supporting the claim that prototype-based readouts are fragile from a single physiological example. One caveat we state rather than leave implicit: the backbone was meta-trained under an adaptation objective, so its features are not optimized for prototype separation.

W1, baseline accounting. We should have stated our external comparisons explicitly rather than leaving them to be read off row labels. Table 1 contains two, and we are relabeling both. *Kaifosh and Reardon [11]* (Nature, 2025) is a pretrained neuromotor decoder trained on more than 4,800 participants, architecturally external to our work and evaluated both as a linear probe and with full fine-tuning. *Côté-Allard-style transfer learning [7]* is the row currently labelled "No MAML, No MoE": a CNN-LSTM pretrained across users and fine-tuned per novel test user under L2SO.

On the second, we argue that pretraining on our own users is the correct comparison here. The contribution of [7] is a transfer-learning procedure for EMG, cross-subject supervised pretraining followed by per-user fine-tuning, not a specific architecture. Running their forearm-band network on our data (or pretraining on their data) would require synthesising channels (their data only had 8 channels of EMG, and 0 of IMU), and would then measure our channel-reduction decision and the electrode-placement shift between a wrist band and an upper-body array rather than the procedure itself. What this does leave is that we compare procedures under our protocol rather than reproducing published numbers, and we will state that.

W1, W2: architecture baselines, and the transformer we did not run. We promised a transformer comparison and did not produce one within the discussion period. We are not going to report an untuned transformer, which would be uninformative whichever way it fell. Instead:

- We are **removing the comparative assertion at L171–173** that recurrent inductive bias is preferable to attention under one-shot inner-loop adaptation. It came from pilot testing and the paper should not assert it; what remains is a stated design decision with its rationale.
- That rationale is now anchored to a measurement rather than a hypothesis. Our MoEMeta Port B result (below) shows that restricting inner-loop adaptation to the classification head costs 28.6 points here, so adaptation of the *features* is necessary in this regime rather than incidental, which makes the degrees of freedom an encoder exposes to a one-example inner loop a genuine design consideration. It does not establish that attention would adapt worse.
- For camera-ready we commit to the controlled comparison with a pre-specified protocol: CNN experts and LSTM replaced by transformer equivalents, matched parameter count, its own HPO budget under the identical search protocol, evaluated on the same L2SO folds, reported whichever way it falls, as we did for Ports A and B.

Results the reviewer may not have seen. Two experiments were completed during the discussion but posted in Reviewer ARaC's thread; we should have consolidated them. *Port A*: support-derived vs. query-conditioned routing, 91.24% vs. 91.25%, a null, on the basis of which we withdrew an accuracy claim from the paper. *Port B*: restricting inner-loop adaptation to the classification head (8,451 of 5,535,609 parameters, 0.15%) costs 28.6 points (59.76% ± 2.98% vs. 88.4%), consistently across all four held-out subjects, with both possible interpretations stated in advance of the run.

We would also note that the subject-level routing analysis in our 30 Jul reply (Q3) was new computation rather than a commitment: gate vectors across all 32 subjects show no subject holds a private expert (largest margin 0.050 against a ceiling of 1.0).

Official Review of Submission31103 by Reviewer xh89

Official Review by Reviewer xh89 26 Jun 2026, 01:43 (modified: 07 Aug 2026, 02:40)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer xh89

Revisions (/revisions?id=IMaoMipx10)

Summary:
EncoderMoE, combines MAML++ (meta-learning) with a Mixture-of-Experts encoder bank for few-shot, cross-subject gesture recognition from multimodal EMG+IMU signals. The paper identifies three challenges (physiological heterogeneity, semantic heterogeneity, few-shot adaptation) and EncoderMoE addresses all three. The MoE encoder (22 expert CNNs) routes samples to specialized sub-encoders based on signal characteristics, while MAML++ learns parameter initializations that enable rapid adaptation to unseen users. Evaluated on 32 participants (26 with motor impairments) performing 10 user-defined gestures, achieving 86.7% on 1-shot 3-way classification.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:

Strength

- Valid and important problem.
- Authors conudcted a set of careful experiment, with parameter-matched, proper statistical testing.

Weakness

- The practical deployment scenario would be 10 gestures (the user's full vocabulary), but the primary evaluation is 3-way classification. The 10-way result (67.7%, Table 4) is much weaker
- MAML for EMG calibration has been done before, e.g,. Proroković et al. 2020
- The ablations are well-designed (parameter-matched, proper statistical testing). But the experiments are limited in scope: Single dataset, single session — no test of cross-session, cross-device, or cross-task generalizability, Only 10 repetitions per gesture per user— real-world variability (fatigue, electrode shift, day-to-day changes) not captured.
- The paper claims multimodal fusion is important but provides zero evidence that IMU adds value over EMG alone (or vice versa).

Quality: 3: good
Clarity: 3: good
Significance: 2: not good
Originality: 2: not good
Questions:

- Foundation models and scaling: How does this compare to the scenario where much larger data is available? The paper compares against "Meta Pretrained EMG" [Kaifosh and Reardon, 2025]. But If paper has 4,800+ users *with motor impairments* performing *user-defined* gestures, would a foundation model approach (scale alone) solve this? Or is the meta-learning + MoE structure specifically needed for high-heterogeneity population
- Cross-subject meta-training outperforms within-subject training, is this highly due to the limited samples per user used for within-subject training?
- How does performance change as you vary the number of meta-training users (e.g., 8, 16, 24, 30)?
- Could authors share in which scenario does the MoE fail?

Limitations:
Yes

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:

No

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Reponse to xh89

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

30 Jul 2026, 11:59

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer xh89

Comment:

1. Foundation models and scaling: would scale alone solve this?

We will add the following to [Discussion]: “The architecture of EncoderMoE is uniquely suited to the challenge presented in this paper of individuals with motor disabilities performing user-defined gestures. Our preliminary results showed that a LOSO model trained across users and tested on one withheld user performed near random chance: while scale should help learn EMG patterns, scale alone cannot learn user-defined gestures where the mapping is unknown for a novel user. Thus, it is possible, but unlikely that simply scaling our dataset (individuals with motor disabilities performing user-defined gestures) and training a foundation model will result in successful zero-shot gesture recognition.

That said, there is evidence in [11] itself that scale does not remove the need for personalization in exactly the population we study. Kaifosh and Reardon report that fine-tuning improved handwriting classification the most for the participants the global model served worst, which is consistent with those participants having signal characteristics poorly represented in the pretraining distribution. Our cohort is selected for precisely that property. A dataset of individuals with motor disabilities performing user-defined gestures would have higher heterogeneity than [11]'s corpus on two axes at once: signal characteristics and gesture execution. So we expect the personalization gap they observe to widen, not close, with scale.

There is also a structural point specific to user-defined labels. In [11], deeper layers increasingly mix participant identity and separate gesture identity, which is the desired behaviour when everyone performs the same gesture for the same label. In our setting two users may perform near-identical movements that mean different things, and different movements that mean the same thing. A global encoder would place those near-identical movements close together in representation space regardless of label, so scaling the encoder does not resolve the ambiguity; only user-conditional adaptation does.”

2. Is the cross-subject advantage just a data-volume effect?

Partly, but not only. A within-subject model trains on one user's data, while a meta-trained model sees 24 users, so volume is certainly part of it. But the parameter-matched single-encoder control (MAML, No MoE) sees the identical 24-user corpus and still reaches 81.7% against EncoderMoE's 86.7%, a 5.0-point gap that data volume cannot explain. Cross-subject meta-training buys both more data and cross-user structure that a single encoder does not exploit. Our ablations separate the second from model capacity, but they do not separate it from data volume, and we will say so explicitly rather than implying the comparison is cleaner than it is.

3. How does performance change with the number of meta-training users?

A sweep is running at 4, 8, 16, and 24 meta-training users and we will report the curve and add it to the appendix. [TODO] The existing endpoints bracket it: one meta-training user (Subject-Specific EncoderMoE, 64.6%) to 24 users (86.7%).

4. In which scenarios does the MoE fail?

We will add the following to Discussion: “There are two clear potential scenarios in which EncoderMoE is expected to underperform compared to other architectures.

First is the scenario where there is no shared structure (i.e., similar movements and body parts used across participants). Since EncoderMoE routing draws on a largely shared expert pool rather than partitioning experts across users, it relies on having some structure shared across individuals and gestures. The single-subject case shows this directly: subject-specific EncoderMoE reaches 64.6% against 72.2% for the subject-specific supervised CNN-LSTM. With no other participants to route across, the mixture is overhead.

Second, at large numbers of gesture classes. At 1-shot, EncoderMoE's advantage over MAML-No-MoE is largest (3.8 points at N=3, 12.1 at N=10), but absolute 10-way performance is poor (67.7%) and additional shots do not recover it (64.4% at 3-shot, 68.5% at 5-shot). We take this as a capacity-versus-data limitation: with 32 participants, the model does not have enough data to learn representations that separate ten user-defined classes, and we expect a larger cohort would be needed before the mixture helps at large N.”

Final update for reviewer xh89

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:48

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer xh89

Comment:

Follow-up: completed results

The three substantive requests from the response have now produced numbers.

Q3: meta-training user sweep. $N \in \{4, 8, 16\}$ meta-training users with validation and test splits held fixed, so the only quantity varying is meta-training set size; 3 subsample seeds per N, fixed training seed and hyperparameters, 1-shot 3-way with 9 query samples.

Meta-training users	Test accuracy
4	82.4% ± 0.5%
8	85.3% ± 1.7%
16	87.9% ± 0.5%

Accuracy increases monotonically, and the gain continues from N=8 to N=16, past the point where within-subject sample scarcity alone would account for the cross-subject advantage. We read this as meta-training benefiting from user *diversity* rather than aggregate sample count, which bears directly on Q2. We did not extend to N=24/30 given the timeline; the endpoints already in the paper bracket the curve (1 meta-training user: 64.6%; 24: 86.7%), and we will complete the upper range for camera-ready.

W3: no evidence that IMU adds value over EMG alone. Correct, and now tested. Channels are zero-masked rather than deleted, so parameter count is unchanged. Fixed 24/4/4 split, 1-shot, four held-out subjects, 500 episodes:

	3-way	5-way	10-way
Fused	87.6	80.3	63.3
EMG-only	88.7	71.8	45.2
IMU-only	90.4	81.5	65.4

The result does not go our way and we report it as such. IMU is the dominant modality in this dataset; EMG-only degrades sharply with vocabulary size (−18.1 pp against fused at 10-way, Holm-adjusted p = .023, all four subjects agreeing); IMU-only is statistically indistinguishable from fused at every vocabulary size (adjusted p = .385 / .631 / .646). **We are removing the claim at L34–38 that hybrid EMG–IMU interfaces overcome complementary failure modes**, since we cannot support it on this evidence. With four test subjects the fusion contrast is underpowered (the smallest detectable effect at 80% power is roughly 6–8 pp), and we will re-run under L2SO for camera-ready.

Q4: where the MoE fails. Beyond the two scenarios in our earlier reply, the discussion produced a third and more concrete one. Freezing the expert bank during inner-loop adaptation and adapting only the classification head (0.15% of parameters, and the natural configuration if one assumed a globally optimised mixture suffices) costs 28.6 points here (59.76% ± 2.98% vs. 88.4% at 1-shot 3-way), consistently across all four held-out subjects. This also strengthens our answer to Q2: the restricted model sees the identical 24-user corpus, so the deficit cannot be a data-volume effect.

Q1: scaling. One further data point. We re-ran the Kaifosh comparison with their own preprocessing applied, as we had supplied input under EncoderMoE's normalization convention. Corrected, their linear probe improves to 66.5% (published 62.1%) and full fine-tuning falls to 49.2% (published 56.0%). The corrected numbers do not change the conclusion we draw in our answer to Q1.

Official Review of Submission31103 by Reviewer ARaC

Official Review by Reviewer ARaC 23 Jun 2026, 20:52 (modified: 04 Aug 2026, 08:58)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Revisions (/revisions?id=yhXBQKOnck)

Summary:
EncoderMoE applies a few shot adaptation technique to the problem of EMG+IMU based gesture recognition for outlier users with motor impairments that yield out of distribution motion patterns. Using an existing dataset of 32 people (26 with upper body motor impairments) performing a set of 10 gestures, the a mixture-of-experts model trained using MAML++ is tuned to a specific user's gesture manifestation using as few as 1 labeled example per gesture class. The comparative value of MAML++, the mixture of experts model architecture, and cross-user pretraining is established via an ablation study showing steep falloff (5%-22% degradation) without each of these components. The paper illuminates that MoE MAML++ pretraining with a multi-user dataset yields more diverse and robust latent representations that allow for effective few-shot adaptation on outlier user gesture motions. The method demonstrates strong performance for one to few-shot adaptation with small (<10) gesture vocabularies, while larger gesture sets remain an open problem.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths: The paper demonstrates strong results in adapting to esoteric user behaviors that are not represented in any of the training data. There is clean motivation for each technical component, MoE to learn diverse features rather than collapsing to a median, MAML++ to learn adaptability. The evaluations on K-shot N-class gesture recognition demonstrate strong results beating pretrained and user-fine-tuned models, with that performance gap falling off for high gesture counts and as K increases. The evaluations are rigorous such that, despite a relatively small dataset size, the effectiveness of the method is well measured against ablations. The authors clearly state limitations, that the method works well in a specific regime of small vocabulary size and one to few-shot adaptation while large vocabularies remain an open problem.

Weaknesses: While the ideas in the paper are interesting and the merit is proven in application, I believe many of the technical ideas in the paper are closely related to already published prior work that is not cited (MoEMeta, Han Wu, NeurIPS 2025). The current framing of the paper's contributions is fuzzy given this, whether the contribution is primarily in the application/use case or the method itself.

Quality: 3: good
Clarity: 3: good
Significance: 2: not good
Originality: 2: not good
Questions:

1. Are the authors aware of the paper "MoEMeta" from Han Wu in NeurIPS 2025? This should definitely be a citation as it feels like it covers most of the methodological contributions claimed in this paper, if perturbed slightly. At least a comparison should be done against MoEMeta to harden the claim that the methodological contributions in the paper carry weight.
2. Does the "Subject-Specific Transfer Learning" model pretrain on the larger dataset of all users (with the leave-two-out validation strategy)? Or is this model trained from scratch using only a single user's data? It would be nice to see an upper bound achieved with fine-tuning your own parameter-matched pre-trained model on all of a test user's training data, if only to understand how close to "ideal" the EncoderMoE model is getting with K-shot adaptation.
3. Is it fair to include an evaluation that fine tunes the model from Kaifosh & Reardon? Given that the IMU features were omitted and the featurization is different, this seems to muddy the story more than it helps. We see this model does poorly, but not how poorly the EncoderMoE model would do with the same featurization. Perhaps an

apples to apples EncoderMoE featurization would strengthen the comparison?

Limitations:
yes

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
no concerns

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Response to Q2, Q3; Update on PortB for ARaC

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

31 Jul 2026, 22:34

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Comment:

Q2: Subject-Specific Transfer Learning.

The condition currently labeled as Subject-Specific Transfer Learning is misleading, as it is better described as a Subject-Specific local model (no cross-training / pretraining). This condition is trained from scratch on k-shots from a single user, then tested on the remaining withheld gesture repetitions from that user (within-user gesture repetition data split). This was conducted independently for all users. We will rename this row to Subject-Specific CNN-LSTM.

That said, our full ablation (No MAML, No MoE), is more aptly named Subject-Specific Transfer Learning, as it is none other than Cote-Allard-style transfer learning [7]: the CNN-LSTM model is pretrained on a set of users and then independently fine-tuned for each novel test user in k-shots. This is evaluated in a L2SO fashion. We will rename this row to “No MAML, No MoE (Cote-Allard-style transfer learning [7])” to emphasize the comparison against an existing method in the field.

We are currently working on running the upper bound as Reviewer ARaC described; this will be completed for camera-ready, and if it finishes before the Aug 3 deadline, we will follow up with the results here.

Q3: the Kaifosh & Reardon [11] comparison.

We agree the current comparison can be improved. We are running modality ablations and will report both the EMG-only and IMU-only EncoderMoE baselines, the former of which will give a direct comparison to Kaifosh which is likewise EMG-only. We are also relabeling the "Meta Pretrained EMG" row to "Kaifosh [11] Pretrained," for the same reason as above. We will report the comparison either way, including if EncoderMoE also does poorly under that featurization.

MoEMeta Port B: frozen vs. adapted mixture.

Mirroring MoEMeta’s approach, we completed a port which meta-trains EncoderMoE with the inner loop restricted to the classification head (8,451 of 5,535,609 trainable parameters (0.15%)), with experts, gate, router and LSTM frozen during per-task adaptation (natively, EncoderMoE updates the entire model during the inner loop). This yields a 1-shot, 3-way accuracy of 59.76% ± 2.98% on held-out test users, compared with 88.4% for full inner-loop adaptation (main EncoderMoE condition). This trend is consistent across all four held-out users in our fixed user split.

We note that this is not an eval-time freeze. The global bank is frozen only inside the inner loop and continues to receive outer-loop meta-gradients throughout meta-training, matching MoEMeta's regime; we verified non-zero outer-loop gradients on the experts, gate and LSTM under the restriction. Per-layer inner step sizes were meta-learned under the restriction rather than inherited from the full-adaptation configuration, so the restricted model is not handicapped by a step size tuned for a different parameter budget.

Thus, it is clear that EncoderMoE’s inner-loop adaptation is necessary in this setting. It is also what our inductive-bias argument predicts: when gesture semantics are user-defined, a globally optimized mixture has no fixed signal-to-label mapping to specialize over, and a terminal head adapted on top of it cannot recover one.

Response to ARaC

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

30 Jul 2026, 12:11

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Comment:

1. Method or application?

Both, principally the method, and narrower than our framing implies. We claim no novelty for MoE, meta-learning, or their combination: MoEMeta establishes the latter, and we add this citation with explanation. Revised text:

“Contribution 1. Method. EncoderMoE: an encoder-level mixture of experts over multimodal EMG-IMU signal whose experts and routing gate are both meta-learned and adapted from a single example per gesture. No individual component is novel; the contribution is the instantiation, motivated by an inductive-bias argument for why routed encoder capacity suits the user-defined gesture regime, where gesture labels are shared across users but their execution is not.

Contribution 2. Empirical Analysis. Parameter-matched ablations against subject-specific and cross-subject baselines, demonstrating that the MAML-MoE combination is necessary in this high-heterogeneity, user-defined regime.”

New Introduction paragraph. “Neither component of this design is new in isolation. Mixture-of-Experts was introduced for conditional computation [20, 8] and has since been applied to time-series [4], including sEMG gesture recognition, where it absorbs inter-session and inter-subject variability [Shabanpour et al., 2025].

Optimization-based meta-learning has been applied to EMG recalibration for prosthetics [17] and stroke rehabilitation [13], and the two have been combined for few-shot relational learning on knowledge graphs [Wu et al., 2025]. In all this prior work, however, the input-to-label mapping is shared: tasks or subjects differ in signal characteristics or in which patterns are well represented, but a common semantics grounds every label, and specialization supplies capacity for the harder regions of that space. User-defined gestures remove this common ground. The label set is fixed and shared, yet each user's execution can be unique and in some cases mutually exclusive with other users' mappings, so no single global signal-to-label mapping is possible and a mixture optimized once globally cannot resolve the inconsistency. This motivates a mixture whose experts and gate are themselves meta-learned and adapted per user."

New Related Work paragraph, MoE subsection. "MoEMeta [Wu et al., 2025] combines a sparsely-gated MoE with meta-learning for few-shot relational learning on knowledge graphs, using globally optimized MLP experts over pretrained entity embeddings to generate a task-level relation prototype, which is then adapted while the expert bank itself remains frozen. Our setting differs in what the mixture must absorb, and consequently in what is adapted. Because gesture semantics are user-defined rather than shared, routing conditions on signal morphology rather than label identity; because the input is raw multimodal signal rather than a pretrained embedding, the mixture acts as a feature extractor at the encoder level rather than as a prototype generator; and because a globally optimized model cannot resolve a per-user mapping, we adapt the experts and gate in the inner loop rather than freezing them."

2 Mechanics.

3-layer 1D CNNs over raw 88×64 windows, not 2-layer MLPs over TransE pairs; every sample routed, not support triplets only; full end-to-end MAML, not 3 projection vectors with the MoE frozen.

3 Benchmark.

MoEMeta cannot be run on our data: it aggregates 1-hop symbolic neighborhoods (sensor series have no entity vocabulary or adjacency); it optimizes candidate ranking (MRR, Hits@k), with no analog in N-way episodic classification; and it specifies no expert family for raw sensor windows, so any port substitutes one and no longer compares against MoEMeta's experts. Comparing its design decisions to ours with data held fixed is well defined; two ports follow, for Experiments.

Port A: support-derived (MoEMeta) vs. query-conditioned routing (ours). 1-shot, 3-way: 91.24% vs. 91.25%. Our own analysis predicts this null: Sec. 4.6 shows gate weights are stable within a user and differ across ability groups, so routing is user-level, and support and query come from the same user. We withdraw the accuracy claim for query-conditioning and will revise Appendix A.6's closing sentence, keeping it as an inference property only: its cost does not scale with support-set size. It does not bear on our K=1 routing-prior claim, which rests on parameter-matched MAML-No-MoE (3.8 pts at N=3, 12.1 at N=10; Table 4).

Port B: frozen vs. adapted mixture. MoEMeta freezes its expert bank at meta-test; we approximate this by meta-training EncoderMoE with the inner loop restricted to the classification head. Run in progress; number to follow during discussion. Stated in advance: if restricted matches full adaptation, our inner-loop claim does not hold here and we drop it; if it underperforms, the mixture must be adapted, not frozen, when the mapping is user-specific. Either way we report it.

Official Comment by Reviewer ARaC

Official Comment by Reviewer ARaC 02 Aug 2026, 10:27

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Comment:

I thank the authors for the additional experiments, in particular Ports A and B. I note the withdrawal of the query-conditioned routing claim on a null result, and the statement of Port B's interpretation in advance of running it. This is good practice and it increases my confidence in the paper's remaining empirical claims.

My remaining concern is structural rather than empirical. The method as revised is MoE combined with MAML++ applied to a new domain. Full inner-loop adaptation is standard MAML; restricting adaptation to the classification head is ANIL (Raghu et al., 2020). Port B is therefore a MAML-versus-ANIL comparison, and its finding, that full adaptation is required where feature reuse does not carry the task, instantiates a known boundary condition from that literature rather than establishing a new one. This is not a criticism of the experiment, which is well controlled. It does mean that I see no further experiment resolves the originality question, including the Q2 upper bound and Q3 modality ablations currently in progress. Both would improve the paper and I would still like to see them, but neither bears on my originality assessment.

The revision that would address my concern is to relabel Contribution 1 as an application and empirical contribution rather than a method contribution, and to state the practitioner case directly. Given MoEMeta's precedent the frozen-mixture configuration is a natural choice in this domain, and it adapts only 0.15% of parameters. Port B establishes that it costs 28.6 points here. That negative result belongs to this paper rather than to MoEMeta, and it is useful independent of whether the configuration that wins is novel.

Replying to Official Comment by Reviewer ARaC

Updated Contributions for Reviewer ARaC

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

03 Aug 2026, 14:42

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Comment:

We appreciate the guidance and have reworked our contributions, provided below:

"Contribution 1. Application. EncoderMoE: an encoder-level mixture of experts over multimodal EMG-IMU signal whose experts and routing gate are both meta-learned and adapted from a single example per gesture. No component is novel in isolation; the contribution is the instantiation, motivated by an inductive-bias argument for why routed encoder capacity suits the user-defined gesture regime where the label set is shared across users but there is no global gesture-to-label mapping.

Contribution 2. Empirical analysis. Parameter-matched ablations against subject-specific and cross-subject baselines, and against reduced variants of EncoderMoE itself, showing that both components of EncoderMoE (an encoder-level mixture of experts, and inner-loop adaptation over that mixture) are required in this high-heterogeneity, user-defined regime."

Experimental details and results for the updated ablations will be added to Experiments/Appendix in the camera-ready version. We will highlight the cost of partial adaptation in our regime given the new ablation results.

Replying to Updated Contributions for Reviewer ARaC

Addendum: Port B framing, and Q3 update

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:34

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ARaC

Comment:

Following the reviewer's 02 Aug comment, we want to close the framing point, not just the contribution labels.

We accept the reading that full inner-loop adaptation is standard MAML, head-only adaptation is ANIL [Raghu et al., ICLR 2020], and Port B is therefore a MAML-versus-ANIL comparison that instantiates a known boundary condition rather than establishing a new one. We will say so in those terms and cite Raghu et al. directly, rather than presenting the boundary as ours.

We are also stating the practitioner case explicitly, as suggested, in the Discussion:

> "Given MoEMeta's precedent, freezing the expert bank and adapting only a terminal head is a natural configuration for a practitioner building a mixture-based few-shot decoder in this domain, and it adapts 0.15% of parameters. In our setting, it costs 28.6 accuracy points (59.76% ± 2.98% vs. 88.4% at 1-shot 3-way, consistent across all four held-out subjects). Feature reuse does not carry the task when gesture semantics are user-defined: a globally optimized mixture has no fixed signal-to-label mapping to specialize over, and a head adapted on top of it cannot recover one. The practical implication is that the mixture must be adapted, not frozen, wherever the input-to-label mapping is user-specific, and that the cheaper configuration is insufficient here."

Q3: Kaifosh comparison. We re-ran their decoder with its own preprocessing applied, having found that we supplied signal-normalized rather than noise-floor-normalized input, which placed their compression nonlinearity in its near-linear regime. Corrected: linear probing **66.5%** (published 62.1%), full fine-tuning **49.2%** (published 56.0%). We report both, including the direction that moves against us.

The matched-featurization control the reviewer asked for (EncoderMoE trained on the same 2 kHz EMG-only inputs), did not complete in time; training at 2 kHz is substantially more expensive than our 20 Hz pipeline and did not fit the discussion window. It is committed for camera-ready and we will report it whether or not it favours us. In the interim, the closest available evidence is our modality ablation: EncoderMoE restricted to EMG reaches 88.7% at 1-shot 3-way on the fixed split, though at our 20 Hz featurization rather than theirs.

Q3, continued: the modality ablation does not go our way. Zero-masking each modality (fixed split, 1-shot, four held-out subjects): fused 87.6 / 80.3 / 63.3, EMG-only 88.7 / 71.8 / 45.2, IMU-only 90.4 / 81.5 / 65.4 at 3-, 5- and 10-way. IMU is dominant; EMG-only collapses as the vocabulary grows; fusion is not distinguishable from IMU alone. We are removing the multimodal complementarity claim at L34-38 accordingly.

Q2. The upper bound did not complete within the discussion period. We understand from the reviewer's comment that it does not bear on the originality assessment, and we are not presenting it as though it does; it will appear in the camera-ready as committed.

Official Review of Submission31103 by Reviewer fkxS

Official Review by Reviewer fkxS 03 Jun 2026, 08:59 (modified: 09 Aug 2026, 08:38)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Revisions (/revisions?id=8tV40UVF9k)

Summary:

The paper presents an approach to tackle multimodal few-shot gesture recognition using EMG and IMU modalities. The approach is based on a MoE paradigm, which receives the concatenation of the sample's data from both modalities and outputs the features that will then be classified by an LSTM and an MLP. Consistent ablation studies have been done.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

- the ablation study is extensive and sound
- anonymized code has been released
- proper statistical significance analysis has been done

Weaknesses

- it is hard to see the real novelty, as the proposed method is composed of well-known submodules and techniques (MoE, MAML++, LSTM) that are compatible with input and output domains
- there is no comparisons with cited related methods, or on different benchmarks
- as a baseline, the authors didn't do experiments using just one of the modalities at a time, not both altogether

Quality: 2: not good

Clarity: 3: good

Significance: 2: not good

Originality: 1: poor
Questions:

- why did you opt to particularly use 1D-CNNs as Experts and an LSTM as Temporal Encoder instead of more advanced Transformer-based architectures?
- are the EMG and IMU samples' values normalized? because the measurements between different or same subject and modalities can greatly vary in both distribution and magnitude
- did you try analyzing the individual contributions of both modalities? performances of EncoderMoE using only EMG or only IMU data could have been interesting to compare

Limitations:
The authors have not compared their method's performance with the performance of other methods on the same benchmarks, which is mainly what prevents me from giving higher ratings.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
In A.3, L455-456, there is a possible violation of anonymity where the authors say "All experiments were conducted on the NOTS cluster at Rice University's Center for Research Computing (CRC), [...]"

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes

Response to fkxs

Official Comment
by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))
30 Jul 2026, 11:56
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Comment:

1. Are the EMG and IMU sample values normalized?

Yes, both, following [28] as stated in the paper. Per trial, each channel is demeaned and each modality block is then divided by a single shared standard deviation computed separately over the 16 EMG and the 72 IMU channels, before concatenation.

Two consequences are worth stating explicitly. Because each block is independently scaled to unit standard deviation, EMG and IMU arrive at comparable magnitude, so neither dominates the shared experts by scale alone. And because the statistic is computed per trial, no normalization statistic is estimated across subjects and no test-subject data informs the normalization of any training sample. This holds for every EncoderMoE condition and every ablation in Tables 1 and 4. We will add a short subsection stating these facts per modality.

2. EMG vs. IMU contribution (also raised by Reviewers Lpuc and xh89)

Copied from response to Lpuc:

We will add the additional evaluation to Experiments; evaluation results will be updated when results come in [TODO].

Here, we zero-mask the channels of each modality separately, and run all three cases (EMG-only, IMU-zero-masked; IMU-only, EMG-zero-masked; both modalities, no zero masking). We are adding a modality ablation on the fixed 24/4/4 split at 1-shot 3-way: fused 88.4%, EMG-only [[FILL: %]], IMU-only [[FILL: %]].

3. Architecture choices are not validated; novelty concerns

Copied from response to Lpuc:

On validation: the original choice came from pilot testing and the reasoning at lines 171-173, which is a hypothesis rather than evidence. We are replacing the LSTM and CNN components with Transformer-based equivalents and will report the comparison. [TODO].

We accept that no individual component is new; we will reframe our contribution because of MoEMeta as seen above. We claim the instantiation and the empirical result: in this specific regime (one labeled example per class, user-defined label semantics, high physiological heterogeneity from users with motor impairments), routed encoder capacity and meta-learned adaptation are jointly necessary, with parameter-matched controls ruling out capacity as the explanation, and routing's benefit concentrates at K=1 and largely disappears by K=5. Contribution 1 has been rewritten to claim that rather than architectural novelty.

"Method. EncoderMoE: an encoder-level mixture of experts over multimodal EMG–IMU signal whose experts and routing gate are both meta-learned and adapted from a single example per gesture. No individual component is novel; the contribution is the instantiation, motivated by an inductive-bias argument for why routed encoder capacity suits the user-defined gesture regime where gesture labels are shared across users but their execution is not."

Official Comment by Reviewer fkxS

Official Comment by Reviewer fkxS 31 Jul 2026, 05:02
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Comment:
Thank you for the clarification. I am satisfied with your reply to point 1, but not to points 2 and 3.

It is unfeasible to run all those experiments in such a short window, and I was not asking for the results to be reported here but in the camera-ready. What I was really asking was about motivating your architectural choices and arguing what the performances would be while using different combinations of EMG and IMU, and analyzing the nature of these data types

Replying to Official Comment by Reviewer fkxS

Response 2 to fkxs

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

31 Jul 2026, 20:18

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Comment:

We thank the reviewer for the clarification, and focus here on the architectural motivation, the nature of the two modalities, and expected single-modality behaviour.

A. Why convolutional experts

We do not claim convolutional experts are intrinsically more accurate than attention-based ones; three considerations drove the choice.

Temporal extent. Our gestures are short (most under 2 s). Within such a window there is little long-range structure for self-attention to exploit that a convolutional stack plus LSTM cannot capture; the long-context regime where transformers dominate is not ours.

Interaction with the adaptation objective. Our target setting is one-shot personalization: zero-shot transfer failed in pilot testing, and requiring more than one demonstration per gesture is a high burden for our user population. Thus, the binding constraint is the inner-loop adaptation step, not outer-loop training accuracy. In the one-shot limit, gradient-based meta-learning favors strong inductive biases and few adapted parameters; second-order inner-loop optimization through a high-capacity attention stack is more expensive and less stable with a single support example per class [Antoniou et al., ICLR 2019]. Convolutional experts encode locality and translation equivariance, matching the short-time amplitude-modulation structure the classical sEMG feature literature exploits (MAV, RMS, zero crossings, wavelets), and keep per-expert parameter count low enough for a mixture to remain tractable under meta-learning.

Precedent. CNN and CNN-LSTM are extensivley validated for sEMG [7, 11], and transformer sEMG decoders report comparable accuracy rather than a decisive advantage at these dataset scales [Rahimian et al., 2021]. We selected based on adaptation stability and compute cost.

B. Nature of the two modalities

sEMG is a stochastic, zero-mean signal whose discriminative information lies in its amplitude envelope and spectral content. It is non-stationary and sensitive to electrode placement, impedance, and fatigue, producing large inter-subject and inter-session shift, but it registers neuromuscular intent, including in low- and no-movement regimes.

IMU is a near-deterministic kinematic measurement concentrated below ~20 Hz, where the waveform itself carries the discriminative signal; gravity provides a stable reference, so inter-subject shift is smaller. Its failure modes are orientation sensitivity and motion artifact, and it carries signal only during overt movement.

The two therefore fail under near-disjoint conditions, which is the argument for dedicated experts over a shared encoder and implies unequal per-class contributions. This matters most in our dataset of user-defined gestures from users with motor disabilities, where some gestures are very low displacement (e.g. craning the neck); routing exists to exploit this asymmetry.

C. Expected single-modality behaviour

Most of our gestures produce overt movement, so drive and kinematics are coupled and the streams largely redundant. We predict: near-parity between EMG-only, IMU-only and multimodal at low N-way and high k-shot; a widening multimodal advantage as N-way increases and k-shot decreases, since redundancy supports coarse but not fine discrimination among similar gestures; IMU-only degrading disproportionately on the low-displacement subset; and EMG-only degrading disproportionately in cross-subject transfer before adaptation. If a single modality matched multimodal everywhere, that is itself informative for wearable design and we will report it as such. We will provide the full ablation across the N-way/k-shot grid for camera-ready.

D. Comparison with existing methods

Table 1 already reports a Cote-Allard-style transfer baseline [7] under our protocol and compares against [Kaifosh et al., 2025]; per other reviews we will relabel those rows explicitly and add an EMG-only EncoderMoE baseline.

Broader comparison faces a validity issue: to our knowledge no published EMG-IMU model accepts a sensor configuration compatible with ours (existing systems are forearm-based; ours spans the upper body), so running one would require discarding or synthesizing channels, measuring our channel-reduction decision rather than the competing method. Regardless, we will report EMGBench [Yang et al., 2024] in the EMG-only configuration, and do not expect to lead it: its gesture sets are standardized, so a single global signal-to-label mapping exists and scale and pretraining should dominate. Our claim is that routed, per-user-adapted capacity is required when no such mapping exists, as when 32 users independently define the execution of a shared label set, sometimes contradictorily. That predicts high expert specialization on our data and collapse toward uniform routing on EMGBench; we will report gate statistics on both, and a null result counts against us. On novelty, please see our response to Reviewer ARaC.

Official Comment by Reviewer fkxS

Official Comment by Reviewer fkxS 02 Aug 2026, 03:02 (modified: 02 Aug 2026, 03:02)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Revisions (/revisions?id=gWPdD4y7wt)

Comment:

- performance of temporal encoders is given by the number of samples per second analyzed in a single pass (in Hz), not the raw length of the record. I am still not convinced about the authors' choice of using LSTMs
- do the authors mean [2] when citing [Antoniou et al., ICLR 2019]? it is not clear, and the only paper matching the author's name in the paper is a preprint
- [Rahimian et al., 2021] has not been cited at all

Regarding the rest, I thank the reviewers for their engagement and clarifications. I find this discussion about the characteristics of the different natures of the used modalities hard to follow and not easily structured, but it makes sense to me

Replying to Official Comment by Reviewer fkxS

Response 3 to Reviewer fkxS

Official Comment

by Authors (Momona Yamagami (/profile?id=~Momona_Yamagami1), Kai Malcolm (/profile?id=~Kai_Malcolm1))

04 Aug 2026, 00:38

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Comment:
Four points, and one result that requires us to retract an argument from our previous reply.

1. Temporal encoders and sampling rate. The reviewer is right that the relevant quantity is what the encoder processes in a single pass, and our appeal to gesture duration in seconds was the wrong framing. The numbers: EMG is acquired at 2,000 Hz but high-pass filtered at 40 Hz, half-wave rectified, low-pass filtered at 40 Hz and **downsampled to 20 Hz** via a 100 ms sliding window with 50 ms overlap; IMU is acquired at 148 Hz, with no further preprocessing. Both modalities are then resampled to **T = 64 time steps** (§4.1, L206–210). The convolutional stack uses same-padding and unit stride, so the LSTM, or any attention replacement, operates over a 64-step sequence.

We should be precise about what this does and does not support. It removes any concern that our input is a long, high-rate sequence, and, equally, any argument that attention would be computationally impractical here. Our concern was never inference cost but adaptation: at meta-test the inner loop must specialize from a single example per class, and the parameter count exposed to that update, together with its stability, is the binding constraint [Antoniou et al., ICLR 2019]. This is not an argument that the encoder is trained from 64-step examples (it is meta-trained across 24 users and 500 episodes per epoch), but that the *adaptation* step operates on that budget.

Our MoEMeta Port B result bears on this: restricting inner-loop adaptation to the classification-head-only costs 28.6 points here (59.76% ± 2.98% vs. 88.4%), so inner-loop adaptation of the *features* is necessary in this regime rather than incidental. That makes the degrees of freedom an encoder exposes to a one-example inner loop a real design consideration. It does not establish that an attention encoder would adapt worse, so we are removing the comparative assertion at L171–173 and retaining the design rationale, with the controlled comparison (transformer experts and temporal encoder, matched parameters, its own HPO budget, the same L2SO folds) pre-specified for camera-ready and reported whichever way it falls.

2. Antoniou. Reference [2] is the same work. Our entry points to the arXiv preprint (arXiv:1810.09502); the canonical version is ICLR 2019 and we are updating the entry and citing it consistently.

3. Rahimian et al., 2021. Correct: cited in our response and absent from the references. It will be added for camera ready.

4. Modality contributions, and a retraction. In our previous reply we predicted, before running this, near-parity at low N-way and a widening multimodal advantage as N-way increased. The first held; the second did not. Fixed 24/4/4 split, 1-shot, four held-out subjects, 500 episodes:

	3-way	5-way	10-way
Fused	87.6	80.3	63.3
EMG-only	88.7	71.8	45.2
IMU-only	90.4	81.5	65.4

A two-way repeated-measures ANOVA gives a significant condition × n-way interaction ($F(4,12) = 19.2$, $p < .001$). EMG-only falls significantly below fused at 10-way ($\Delta = -18.1$ pp, Holm-adjusted $p = .023$, all four subjects agreeing) and marginally at 5-way ($\Delta = -8.5$ pp, adjusted $p = .076$). IMU-only is indistinguishable from fused at every vocabulary size (adjusted $p = .385$ / $.631$ / $.646$).

So the prediction that the modalities are not interchangeable held, since EMG-only collapses as the vocabulary grows, but the prediction that fusion would pull ahead did not. **We are removing the claim at L34–38 that hybrid EMG–IMU interfaces overcome complementary failure modes, and replacing it with what we can support: IMU is the dominant modality in this dataset, EMG alone degrades sharply as the label set grows, and we do not claim fusion improves over IMU alone on this split.** With four test subjects the fusion contrast is underpowered (the smallest detectable effect at 80% power is roughly 6–8 pp), and we will re-run under L2SO for camera-ready so it enters the same paired design as our main results.

One observation relevant to the reviewer's question about the nature of the two data types: our pipeline reduces EMG to a rectified 20 Hz envelope over 16 channels while IMU retains 72 channels of kinematics, so this result characterizes our EMG *representation* and a 4.5× channel asymmetry rather than sEMG as a modality.

Replying to Response 3 to Reviewer fkxS

Official Comment by Reviewer fkxS

Official Comment by Reviewer fkxS 04 Aug 2026, 03:15

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer fkxS

Comment:
This reply did not satisfy me, particularly not citing **here** [Rahimian et al., 2021]. The retraction made by the authors)without being prompted by me) and the discussion with the other reviewers makes hard to trust the results obtained by the authors

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